HAIRUL AZHAR BIN ABU BAKAR

FORWARD-DEPLOYED AI ENGINEER · AI SOLUTIONS ARCHITECT · FOUNDER

Location Singapore citizen, open to relocation **Phone** +65 8014-8456 **Email** hairul.azhar.ofc@gmail.com
Portfolio hairulazhar.edgeone.app **LinkedIn** linkedin.com/in/hairul-azhar-7095b9347
GitHub github.com/CybergigSG

Forward-deployed AI engineer and founder who builds agentic AI and runs it in production. I architect **KatalockAi**, a live multilingual AI workspace for India, Singapore and Indonesia; its **523-lesson Academy**; and **AIM OS**, a governed agent platform. I have shipped live client AI products on KatalockAi. Before AI, I spent 10+ years owning budgets, SLAs and on-site delivery in resort fit-outs, hospitality supply and enterprise logistics, so I build AI that operators can trust: verified, gated and recoverable.

LIVE PRODUCTS & CLIENT DELIVERY

KatalockAi
www.katalock.net
Agentic workspace: K-Computer verification, gated browser automation, code sandbox.

KatalockAi Academy
www.katalock.net/academy
523 lessons, 20 phases; AI tutor reviews and grades student code.

AIM OS
aimglobal.io
Governed agent OS: 50+ apps, 108 endpoints, QGov and killswitch.

Tarot with Shilpa · client AI product
www.tarotwithshilpa.com
Reader personas, multilingual readings, freemium metering, human hand-off.

Ignis Eco · client website
www.igniseco.com
Investor site for a plastic-to-fuel venture, with a language switcher.

TECHNICAL SKILLS

AI & agents
Agentic AI · Multi-agent orchestration · Cross-model verification · RAG · MCP · Knowledge distillation · Quantization · Evaluation · Pinecone / Chroma

Full stack & infra
Python · TypeScript · React / Next.js · Node.js · FastAPI / Django · PostgreSQL / Supabase · Redis · Docker / Kubernetes · AWS / GCP · Terraform

PROFESSIONAL EXPERIENCE

AI Platform Architect & Co-Founder | KatalockAi
2026 – Present
Global platform · Remote

- Own backend architecture, agent orchestration and infrastructure end to end; co-founder runs on-the-ground operations.
- Built a tiered model stack (lucy, tara, stargazer, aim) using **knowledge distillation and cost-performance routing**, localised for India, Singapore and Indonesia.
- Shipped K-Computer cross-model verification, browser automation that pauses before any payment, an evidence-ledger CRM, and a code sandbox that publishes and pushes to GitHub.
- Delivered client AI products, including **Tarot with Shilpa** (persona-based readings, freemium metering, human hand-off) and the **Ignis Eco** investor site, plus SME bundles for clinics, retail, real estate and restaurants.

Architect & Founder | KatalockAi Academy
2026 – Present
Global platform · Remote

- Built a **20-phase, 523-lesson** AI engineering platform with 6 career routes, 17 certified capstones and certificates carrying verifiable IDs.
- Built **Professor Katalock**, a Socratic AI tutor that reviews student code, flags edge cases and grades projects against rubrics.
- Curriculum covers LLM serving (vLLM, SGLang, TensorRT-LLM), quantization (AWQ, GPTQ, FP8), alignment (RLHF, DPO, Constitutional AI), MCP and multi-agent systems.

Founder & AI Platform Architect | AIM OS (aimglobal.io) Jan 2023 – Present

Singapore · Remote

- Architected an agent OS spanning 50+ apps, 108 API endpoints, 27+ agents and about 80 tables.
- Designed **QGov** policy engine, **Sentinel** safety monitor and a founder killswitch; built **Nexus Hub** orchestration with intent routing, pg-boss retries and Yjs CRDT collaboration.

Co-Founder, Investor Relations & GTM | Ignis Eco 2026 – Present

Tamil Nadu / Bengaluru, India

- Lead investor relations, seed-stage capital structuring and go-to-market for a plastic-to-fuel pyrolysis venture.

Logistics Technology Operations (contract) | GOGOVAN Singapore (GOGOX) Jul 2024 – Sep 2024

- Ran enterprise logistics technology for IKEA Singapore: **99.8% on-time** across 5,000+ daily orders; incident resolution 25% faster.

Co-Founder & Operations Director | Freda'D Pte Ltd / DGCC Investment Apr 2014 – Dec 2024

Singapore / Middle East · run concurrently with employment roles

- Grew a bootstrapped fragrance business from $500 to a profitable international supplier to luxury hotels, at **25% YoY growth**; automation cut overhead 30%.

Project Director | HH Design / Interior Producer Pte Ltd Jan 2010 – Mar 2020

- Delivered integrated-resort fit-outs for **Marina Bay Sands Casino** and **Resorts World Sentosa** on $500K–$5M budgets, improving timelines 15%.

FEATURED PROJECTS

WorkflowGenius · Enterprise Document Automation github.com/CybergigSG/WorkflowGenius

Python, RAG, Pinecone, React, AWS Lambda/EC2

RAG pipeline processing 10,000+ daily transactions at 99.5% uptime on AWS, reducing administrative workload by 40%.

EDUCATION

MSc Computer Science
Heriot-Watt University · expected May 2028

Diploma in Architectural Technology
Singapore Polytechnic · 2001–2004

CERTIFICATIONS

IBM Full Stack Software Developer
Professional Certificate, 2024

Google AI & Machine Learning
Professional Certificate, 2024

Google Cybersecurity
Professional Certificate, 2024

ADDITIONAL INFORMATION

Languages: English (native), Malay (native) · **Availability:** immediate, or two weeks' notice

Work authorisation: Singapore citizen. Open to relocation: USA, UK, EU, UAE, Australia, Canada, India, APAC.